Alpha Beta Threshold Selection Methods Study

HELIX FORGE LAB / p04-full-002005 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded threshold selection evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000015 (models/model-cards/model-reviews/alpha-beta/calibration/tables/alpha-beta-calibration-metrics-v01.csv).

Slice	Records	Precision	Recall	F1
slice-01	230	0.862	0.880	0.871
slice-02	184	0.879	0.856	0.867
slice-03	227	0.896	0.869	0.882
slice-04	220	0.878	0.882	0.880
slice-05	187	0.895	0.858	0.876